Toxicokinetic Chemical Simulator (ToCS): A Graphical User Interface for High-Throughput Chemical Analysis

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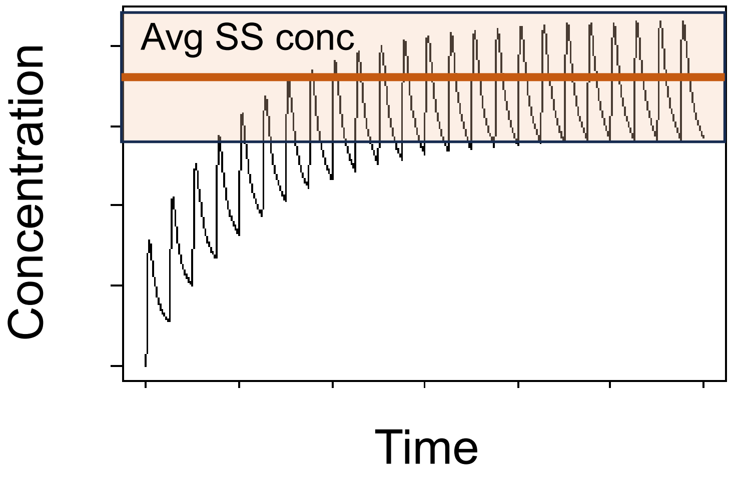
Office of Laboratory Operations and Applied Science (OLOAS)

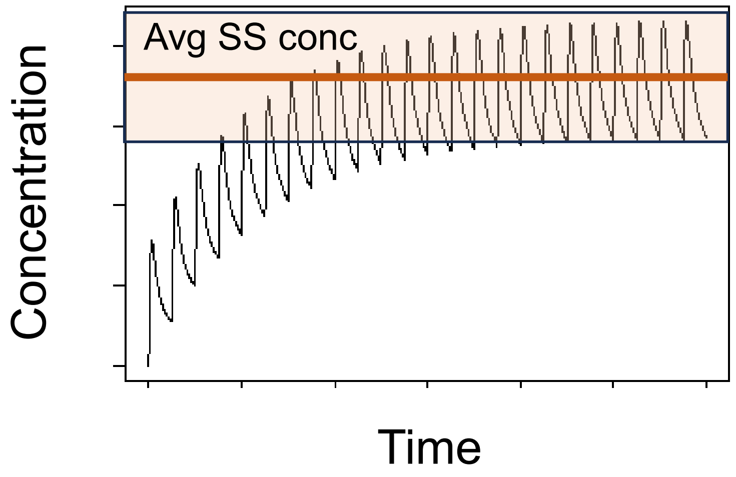
Session IV:
Steady State Concentrations and Parameter Calculations Modules

Steady State Concentrations Module Overview



- A chemical's steady state (SS) concentration is its concentration when the rate of chemical entering and leaving the body are equal during repeated exposures
- The exposure level ("dose"), clearance rate, and bioavailability of a chemical all play a role in determining SS
- Useful in knowing how quickly your chemical accumulates during repeated exposures

Steady State Concentrations Module Outputs



1. SS concentrations table and plot for specified dosing amount during a “constant” oral infusion (every hour for 24 hours)
   - Calculated by a single analytic (algebraic) equation
2. A table with an estimate on the number of days it takes for your chemical to reach SS
   - Calculated by comparing the solution above to numerically solving the model for 3x daily oral dosing. When the model reaches SS value from above or when its fractional change is VERY small, then the number of days is recorded.

Steady State Concentrations Module Inputs

Required:

- Species and species preferences (if applicable)
- Model selection
- In silico parameter preference
- Dosing amount
- Chemicals to simulate

Optional:

- Bioavailability settings
- Output units
- Output units, concentration type (plasma, blood), and tissue compartment
- Model calculation preferences for hepatic clearance, fraction unbound, and partition coefficients

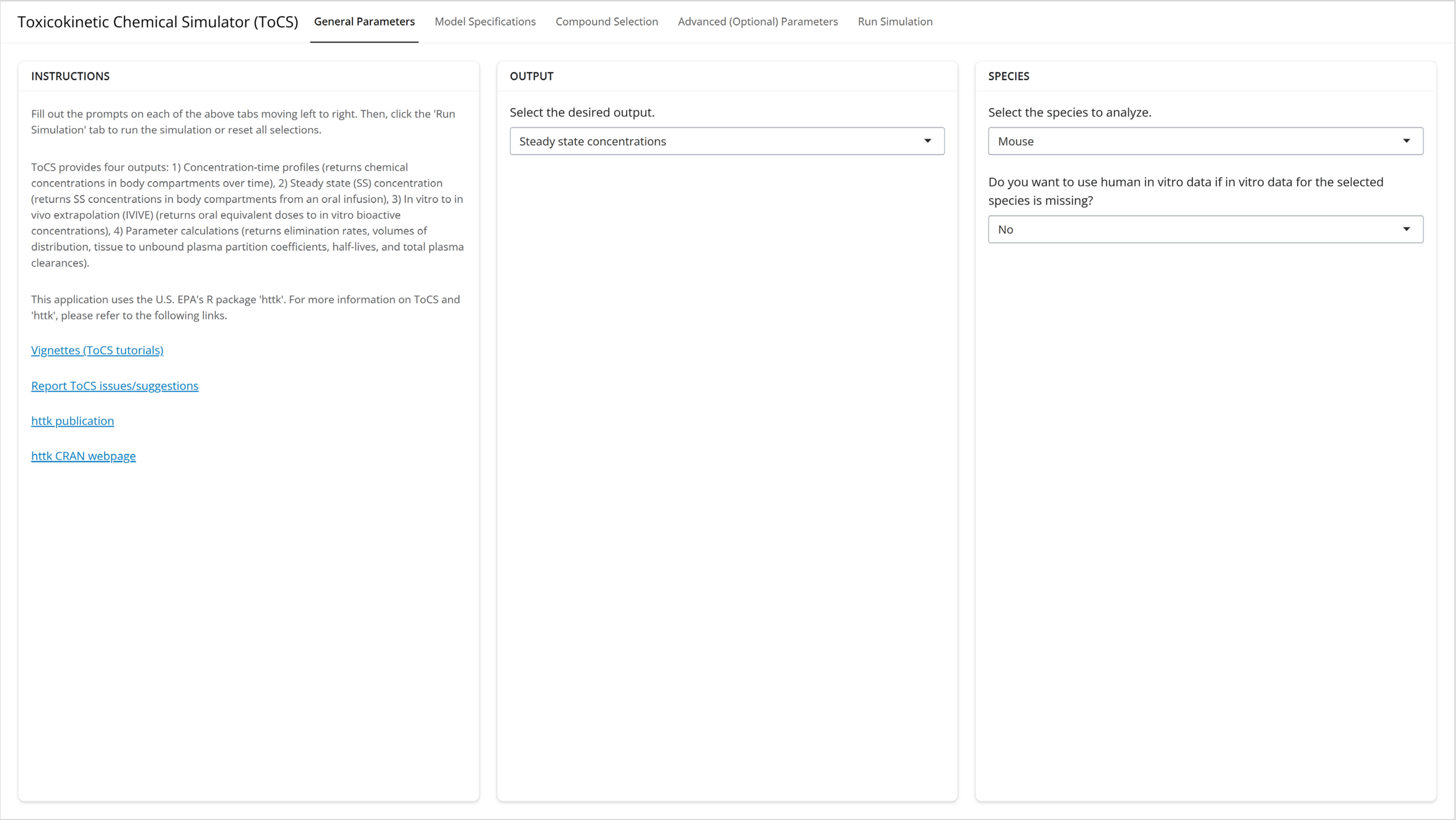
Example 1 Scenario

- Goal:
  - Obtain the mouse steady state concentrations and characteristics of seven food colorings using the pbtk model at 0.5 mg/kg dose per day

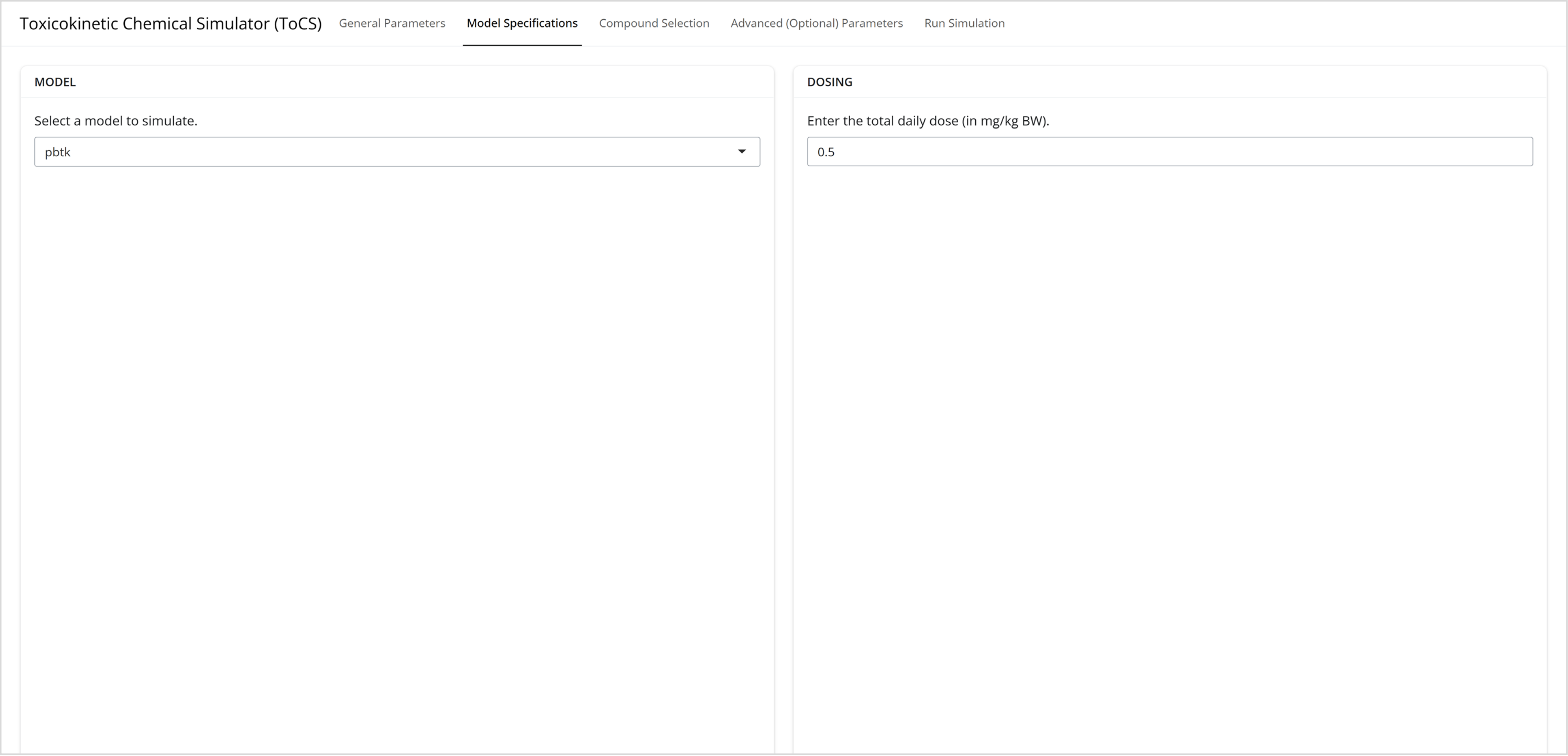
Chemical Name	CASRN
FD&C Red 4	4548-53-2
FD&C Yellow 6	2783-94-0
FD&C Blue 1	3844-45-9
FD&C Yellow 5	1934-21-0
FD&C Red 3	16423-68-0
FD&C Green 3	2353-45-9
FD&C Green 1	4680-78-8

  - Use only mouse data if able
  - Do not use the available in silico parameters
  - Use 100% bioavailability

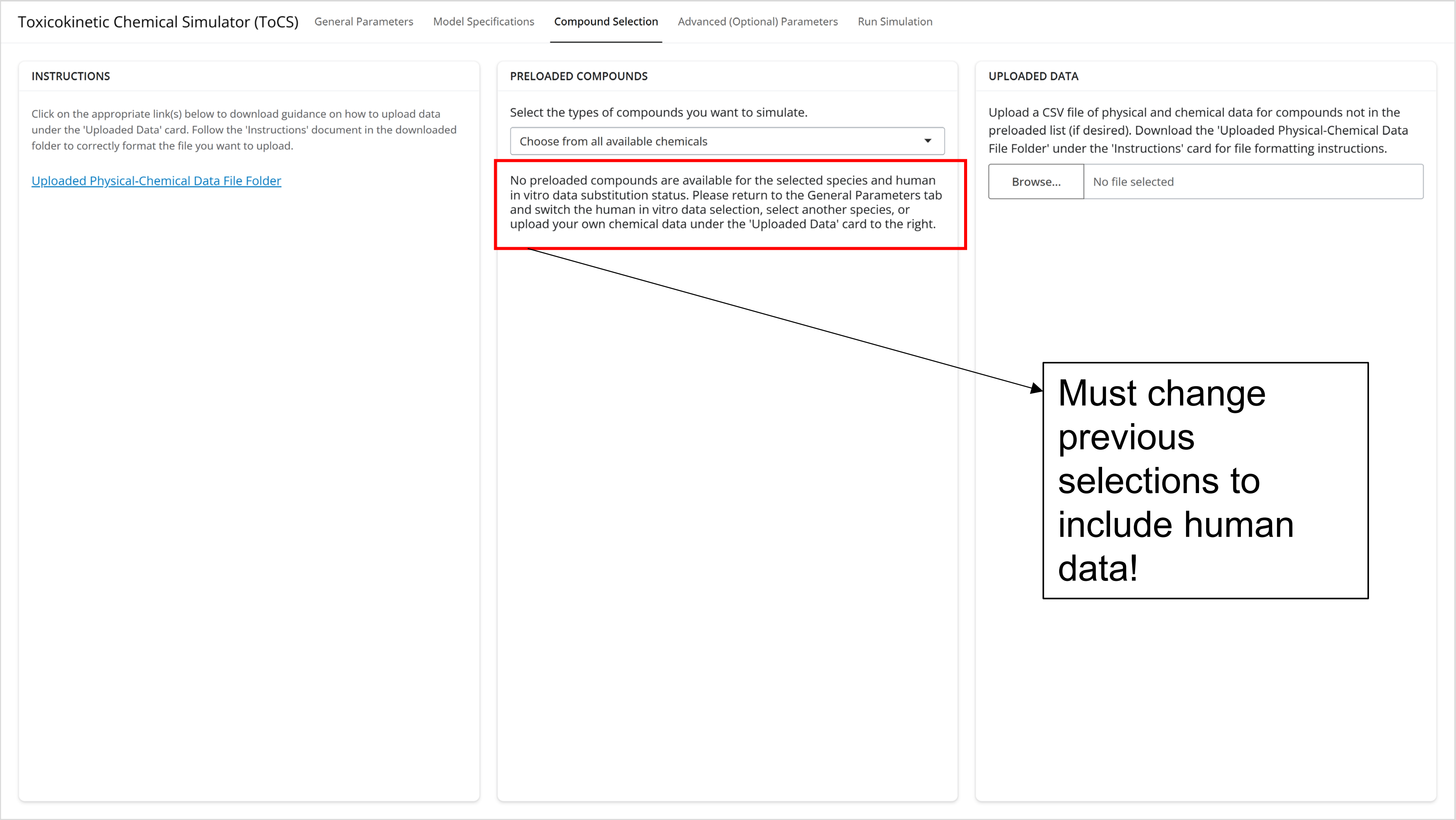
Example 1: General Parameters Tab



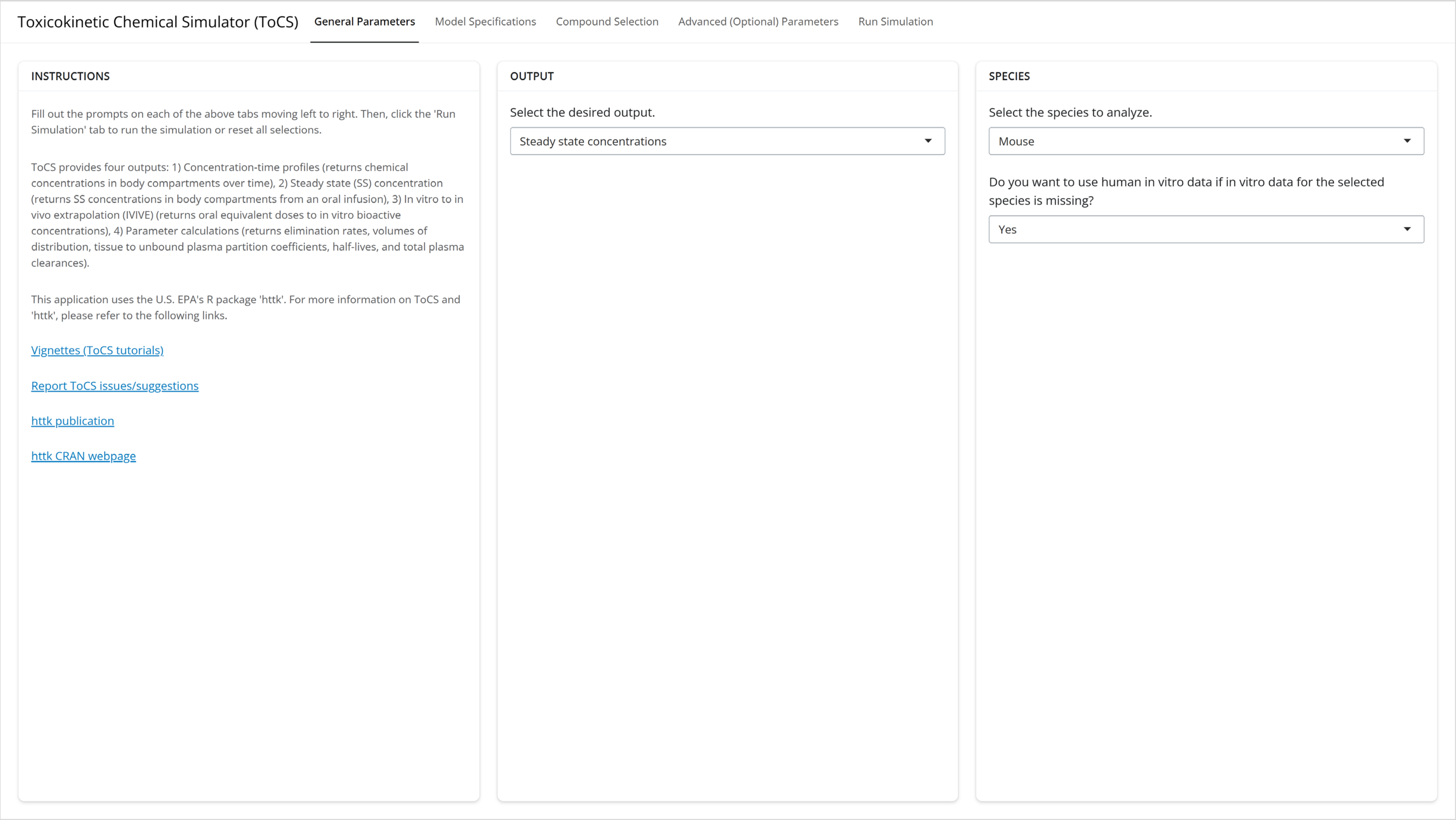
Example 1: Model Specifications Tab



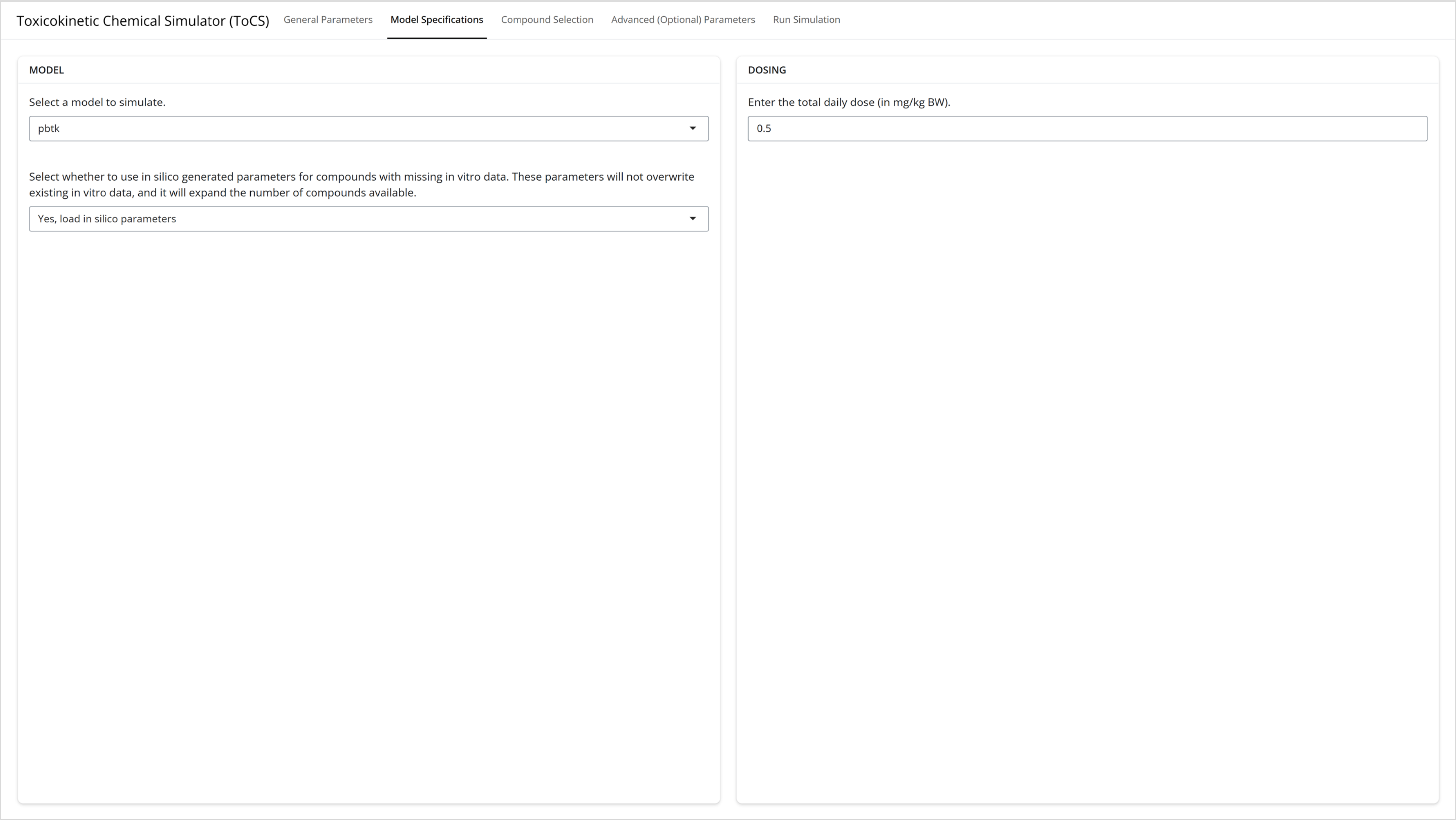
Example 1: Compound Selection Tab



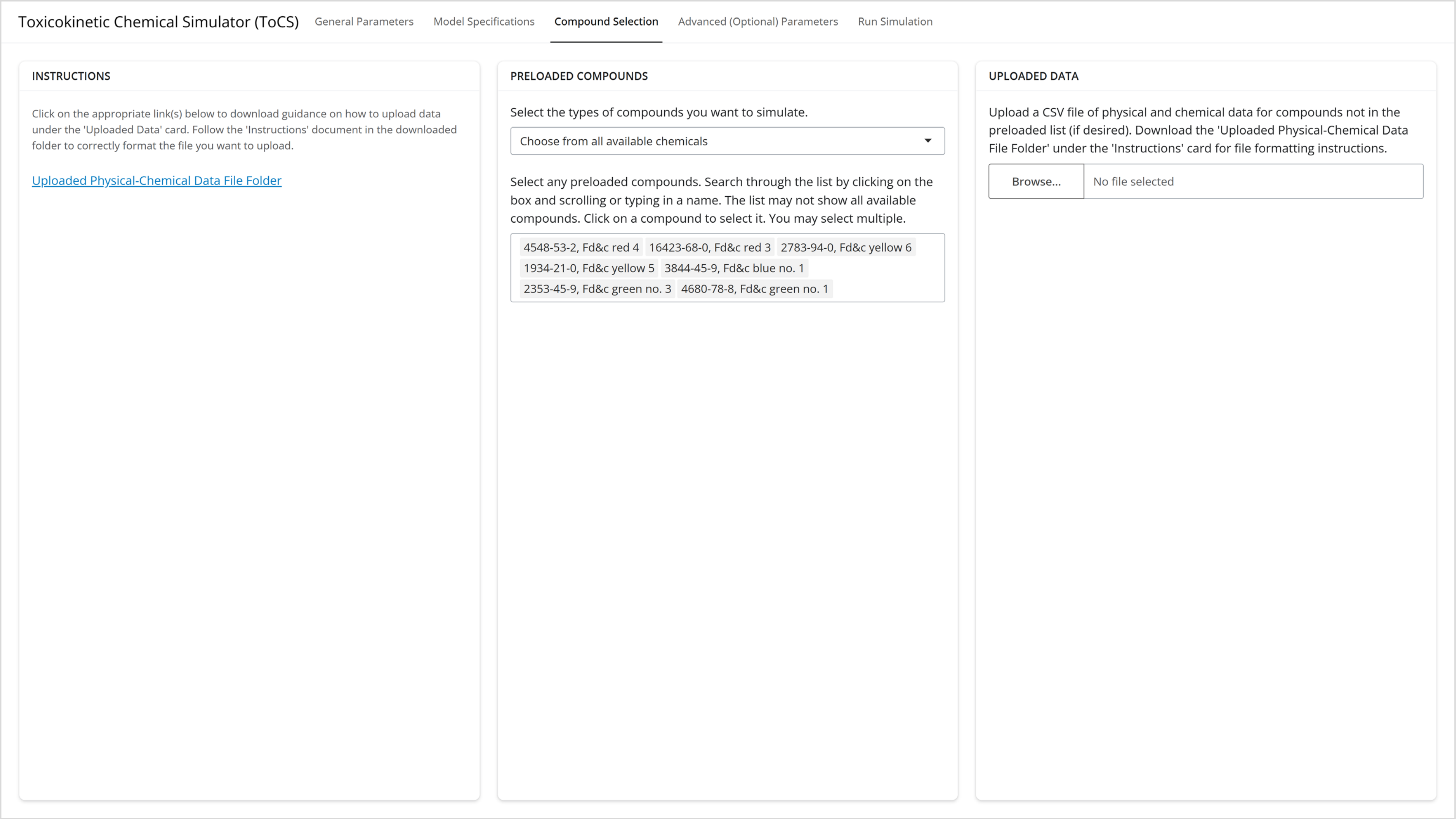
Example 1: General Parameters Tab



Example 1: Model Specifications Tab



Example 1: Compound Selection Tab



Example 1: Advanced Parameters Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | **Advanced (Optional) Parameters** | Run Simulation

MODEL CONDITIONS

Select whether protein binding is taken into account in liver clearance.

Yes, include protein binding (default)

Select whether to adjust the chemical fraction unbound in presence of plasma proteins for lipid binding.

Yes, adjust the fraction of unbound plasma (default)

Select whether to use regressions when calculating partition coefficients.

Use regressions (default)

Enter the p-value threshold for the in vitro intrinsic hepatic clearance rate where clearance assay results with p-values above this threshold are set to zero.

0.05

Enter the minimum acceptable chemical fraction unbound in presence of plasma proteins. All values below this will be set to this value.

0.0001

MODEL SOLVER

No options for this category.

BIOAVAILABILITY

Enter a default value for the Caco-2 apical-to-basal membrane permeability (denoted Caco2.Pab, 10^-6 cm/s).

1.6

Select whether to use the Caco2.Pab value set above to estimate F_abs (the in vivo measured fraction of an oral dose absorbed from the gut lumen into the gut) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default)

Select whether to use the Caco2.Pab value set above to calculate F_gut (the in vivo measured fraction of an oral dose that passes gut metabolism and clearance) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default)

Select whether to overwrite in vivo F_abs and F_gut data (if available).

Do not overwrite in vivo values (default)

Select whether to keep F_abs and F_gut at 100% availability (which overwrites all other bioavailability parameter settings above).

Keep Fabs and Fgut at 100% availability

OUTPUT SPECIFICATION

Select the output concentration units.

uM

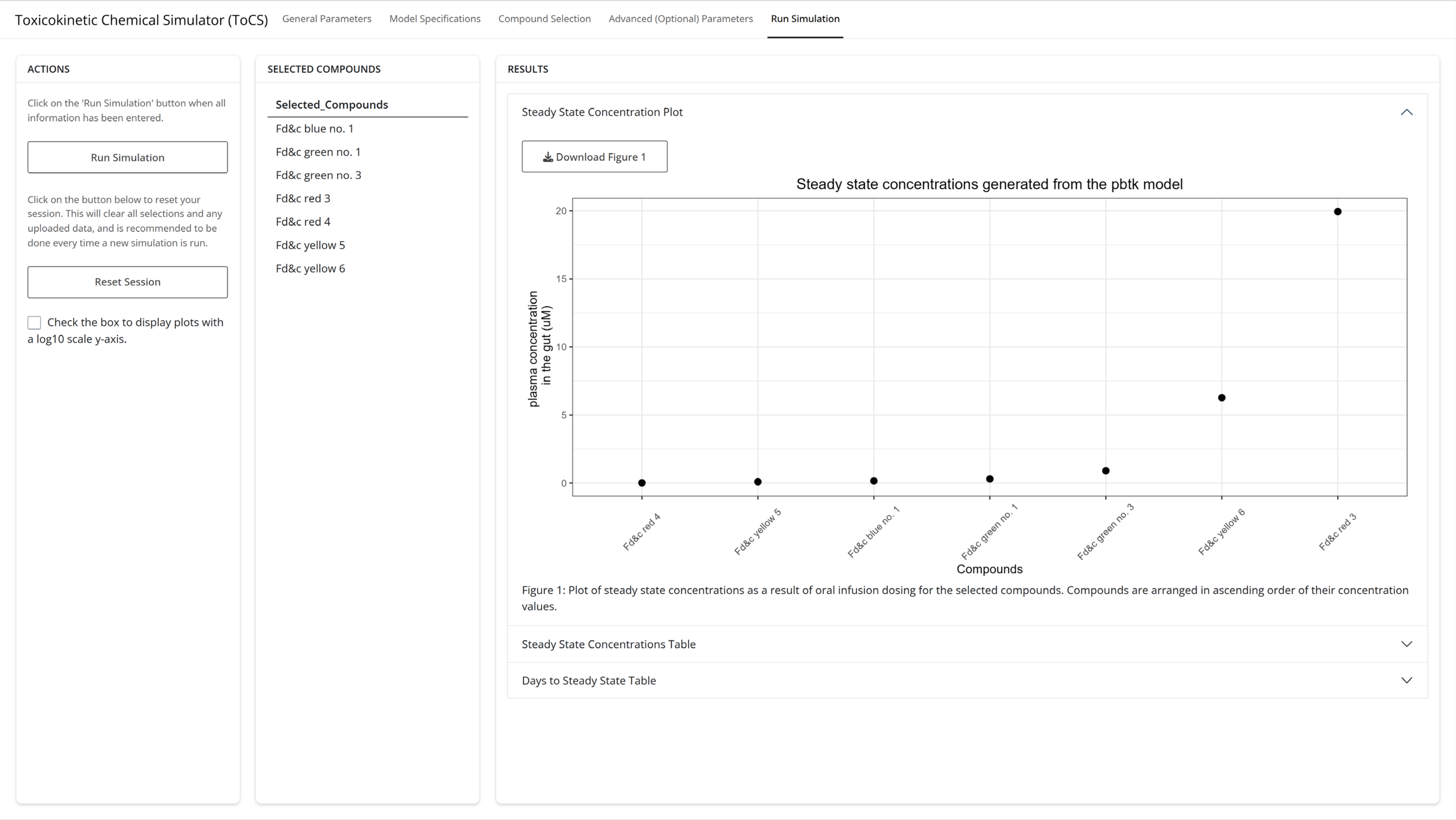
Select the output concentration type.

plasma

Select a tissue you want the output concentration in. Leave on 'NULL' if the whole body concentration is desired.

gut

Example 1: Run Simulation Tab



Example 1: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | **Run Simulation**

ACTIONS

Click on the 'Run Simulation' button when all information has been entered.

Run Simulation

Click on the button below to reset your session. This will clear all selections and any uploaded data, and is recommended to be done every time a new simulation is run.

Reset Session

☑ Check the box to display plots with a log10 scale y-axis.

SELECTED COMPOUNDS

Selected_Compounds
Fd&c blue no. 1
Fd&c green no. 1
Fd&c green no. 3
Fd&c red 3
Fd&c red 4
Fd&c yellow 5
Fd&c yellow 6

RESULTS

Steady State Concentration Plot

Download Figure 1

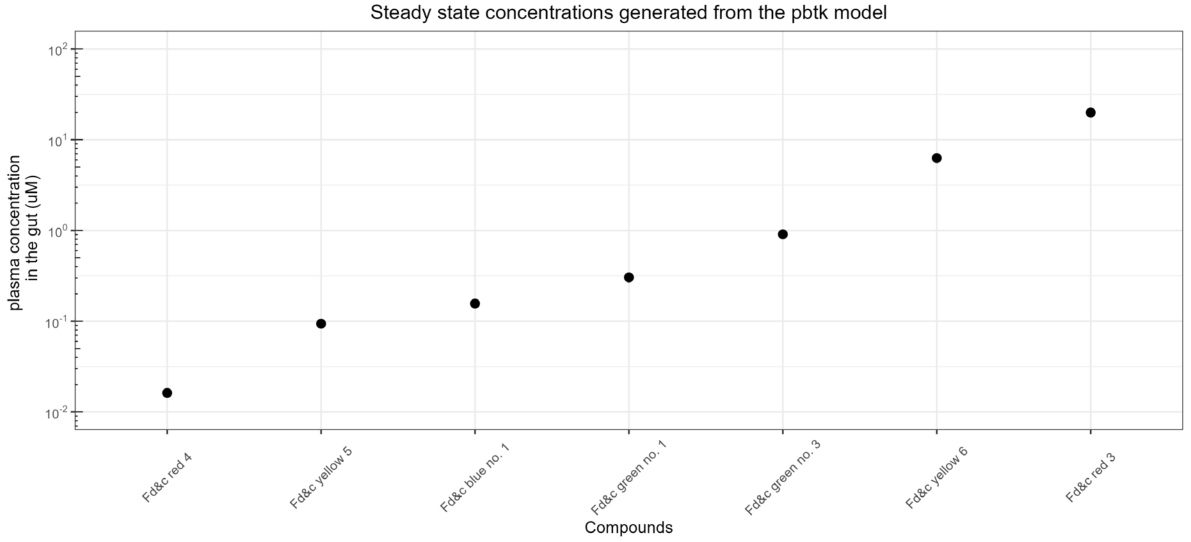


Figure 1: Plot of steady state concentrations as a result of oral infusion dosing for the selected compounds. Compounds are arranged in ascending order of their concentration values.

Steady State Concentrations Table

Days to Steady State Table

Example 1: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | Run Simulation

ACTIONS

Click on the 'Run Simulation' button when all information has been entered.

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SELECTED COMPOUNDS

Selected_Compounds
Fd&c blue no. 1
Fd&c green no. 1
Fd&c green no. 3
Fd&c red 3
Fd&c red 4
Fd&c yellow 5
Fd&c yellow 6

RESULTS

Steady State Concentration Plot

Steady State Concentrations Table

Download Table 1 | Download Simulation Parameters

Show 10 entries

Search:

	CompoundName	SteadyState
1	Fd&c red 4	0.0162
2	Fd&c yellow 5	0.0939
3	Fd&c blue no. 1	0.1564
4	Fd&c green no. 1	0.3038
5	Fd&c green no. 3	0.9078
6	Fd&c yellow 6	6.275
7	Fd&c red 3	19.95

Showing 1 to 7 of 7 entries

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Table 1: Table of the steady state concentrations (uM) as a result of oral infusion dosing for the selected compounds in the selected compartment. Compounds are arranged in ascending order of their concentration values.

Days to Steady State Table

Example 1: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | **Run Simulation**

ACTIONS

Click on the 'Run Simulation' button when all information has been entered.

Run Simulation

Click on the button below to reset your session. This will clear all selections and any uploaded data, and is recommended to be done every time a new simulation is run.

Reset Session

☐ Check the box to display plots with a log10 scale y-axis.

SELECTED COMPOUNDS

Selected_Compounds
Fd&c blue no. 1
Fd&c green no. 1
Fd&c green no. 3
Fd&c red 3
Fd&c red 4
Fd&c yellow 5
Fd&c yellow 6

RESULTS

Steady State Concentration Plot

Steady State Concentrations Table

Days to Steady State Table

Download Table 2

Show 10 entries

Search:

	CompoundName	CssDay	AvgConc	RatioAvgAnalytical	MaxConc
1	Fd&c red 4	0	0.06358	0.9005	0.4299
2	Fd&c yellow 5	1	0.3389	0.9334	0.7727
3	Fd&c blue no. 1	3	0.258	0.9556	0.4268
4	Fd&c green no. 1	6	0.7717	0.9704	1.014
5	Fd&c green no. 3	12	2.018	0.9912	2.2
6	Fd&c yellow 6	109	57.28	0.9989	57.92
7	Fd&c red 3	125	4.805	0.9984	4.838

Showing 1 to 7 of 7 entries

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Table 2: Table of steady state (SS) characteristics. CssDay represents the number of days it takes for the model to reach the analytical plasma SS concentration or the fractional change of daily SS plasma concentration is below the set threshold, AvgConc represents the average plasma concentration (uM) on the final day of the simulation, RatioAvgAnalytical represents the fraction of the analytical SS plasma concentration reached on CssDay, and MaxConc is the maximum plasma concentration (uM) of the simulation.

Parameter Calculations Module Overview and Outputs

- Reasons for Use
  - Gain insight into common TK parameters/attributes of chemicals
  - Compare with in vivo data
  - Input into other mathematical models

- Parameter Outputs:
  - Elimination rate
  - Half-life
  - Total plasma clearance
  - Volume of distribution
  - Partition coefficients for 13 tissues

Parameter Calculations Module Inputs

Required:

- Species and species preferences (if applicable)
- In silico parameter preference
- Chemicals to simulate

Optional:

- Bioavailability settings
- Model calculation preferences for hepatic clearance, fraction unbound, and partition coefficients

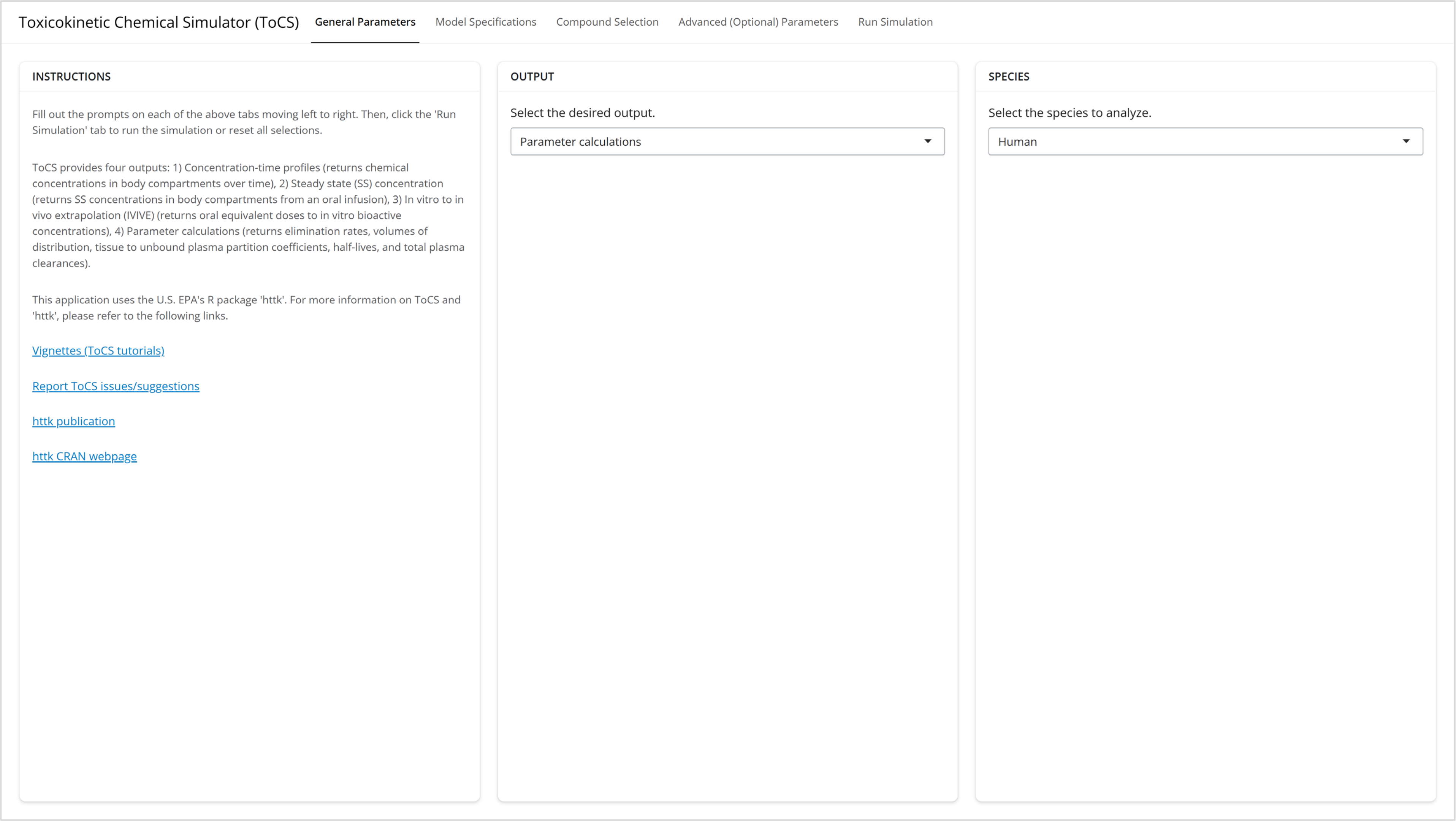
Example 1 Scenario

- Goal:
  - Obtain the human TK parameter values of seven food colorings

Chemical Name	CASRN
FD&C Red 4	4548-53-2
FD&C Yellow 6	2783-94-0
FD&C Blue 1	3844-45-9
FD&C Yellow 5	1934-21-0
FD&C Red 3	16423-68-0
FD&C Green 3	2353-45-9
FD&C Green 1	4680-78-8

  - Use the available in silico parameters

Example 1: General Parameters Tab



Example 1: Model Specifications Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | **Model Specifications** | Compound Selection | Advanced (Optional) Parameters | Run Simulation

MODEL

Select a model to simulate.

Schmitt ▾

Select whether to use in silico generated parameters for compounds with missing in vitro data. These parameters will not overwrite existing in vitro data, and it will expand the number of compounds available.

Yes, load in silico parameters ▾

DOSING

No options for this category.

Example 1: Compound Selection Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | **Compound Selection** | Advanced (Optional) Parameters | Run Simulation

INSTRUCTIONS

Click on the appropriate link(s) below to download guidance on how to upload data under the 'Uploaded Data' card. Follow the 'Instructions' document in the downloaded folder to correctly format the file you want to upload.

Uploaded Physical-Chemical Data File Folder

PRELOADED COMPOUNDS

Select the types of compounds you want to simulate.

Choose from only food relevant chemicals

Select any preloaded compounds. Search through the list by clicking on the box and scrolling or typing in a name. The list may not show all available compounds. Click on a compound to select it. You may select multiple.

4548-53-2, Fd&c red 4
2783-94-0, Fd&c yellow 6
3844-45-9, Fd&c blue no. 1
1934-21-0, Fd&c yellow 5
16423-68-0, Fd&c red 3
2353-45-9, Fd&c green no. 3
4680-78-8, Fd&c green no. 1

UPLOADED DATA

Upload a CSV file of physical and chemical data for compounds not in the preloaded list (if desired). Download the 'Uploaded Physical-Chemical Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse... | No file selected

Example 1: Advanced Parameters Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | Run Simulation

MODEL CONDITIONS

Select whether protein binding is taken into account in liver clearance.

Yes, include protein binding (default)

Select whether to adjust the chemical fraction unbound in presence of plasma proteins for lipid binding.

Yes, adjust the fraction of unbound plasma (default)

Select whether to use regressions when calculating partition coefficients.

Use regressions (default)

Enter the p-value threshold for the in vitro intrinsic hepatic clearance rate where clearance assay results with p-values above this threshold are set to zero.

0.05

Enter the minimum acceptable chemical fraction unbound in presence of plasma proteins. All values below this will be set to this value.

0.0001

MODEL SOLVER

No options for this category.

BIOAVAILABILITY

Enter a default value for the Caco-2 apical-to-basal membrane permeability (denoted Caco2.Pab, 10^-6 cm/s).

1.6

Select whether to use the Caco2.Pab value set above to estimate F_abs (the in vivo measured fraction of an oral dose absorbed from the gut lumen into the gut) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default)

Select whether to use the Caco2.Pab value set above to calculate F_gut (the in vivo measured fraction of an oral dose that passes gut metabolism and clearance) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default)

Select whether to overwrite in vivo F_abs and F_gut data (if available).

Do not overwrite in vivo values (default)

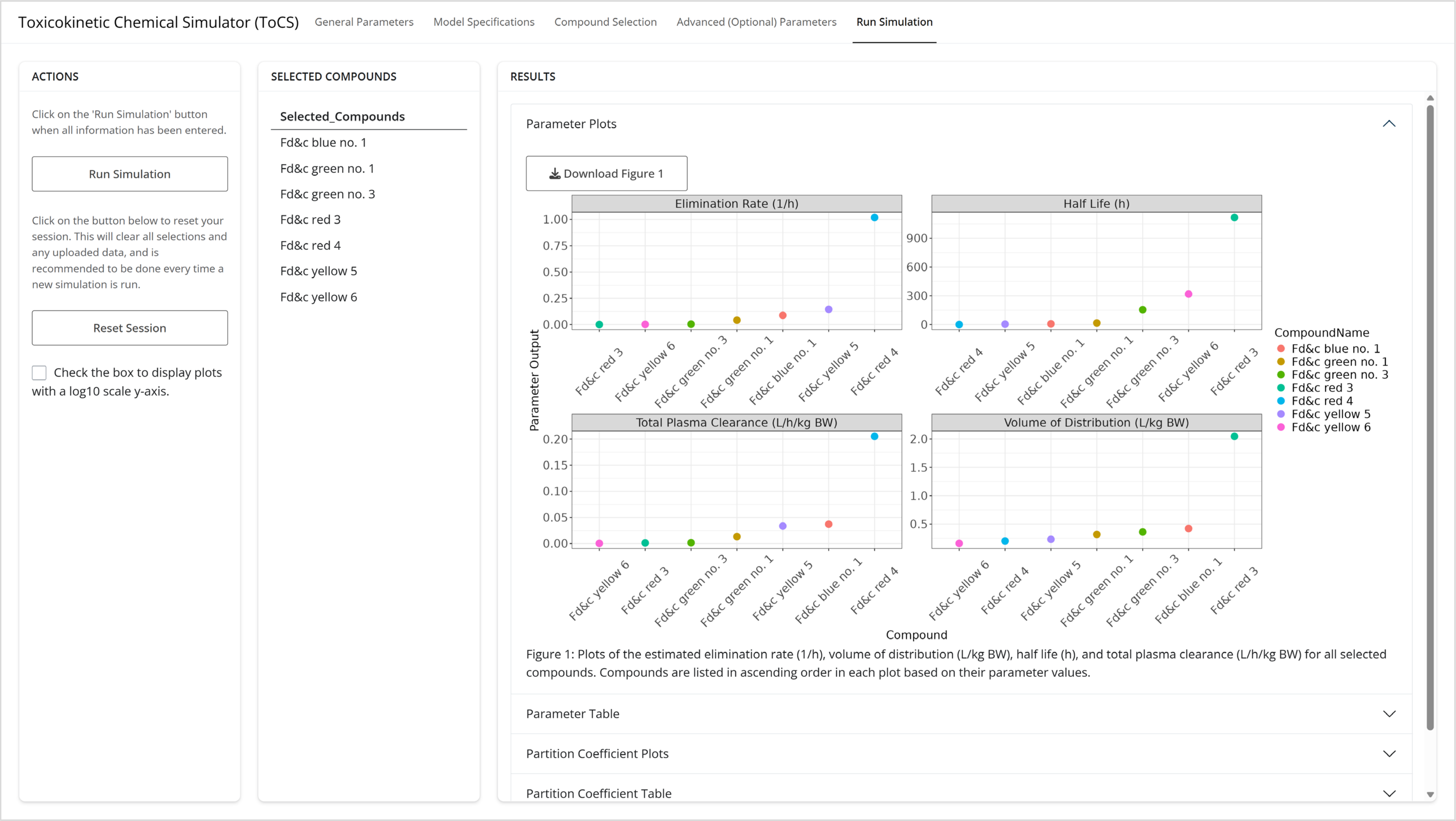
Select whether to keep F_abs and F_gut at 100% availability (which overwrites all other bioavailability parameter settings above).

Do not keep Fabs and Fgut at 100% availability (default)

OUTPUT SPECIFICATION

No options for this category.

Example 1: Run Simulation Tab



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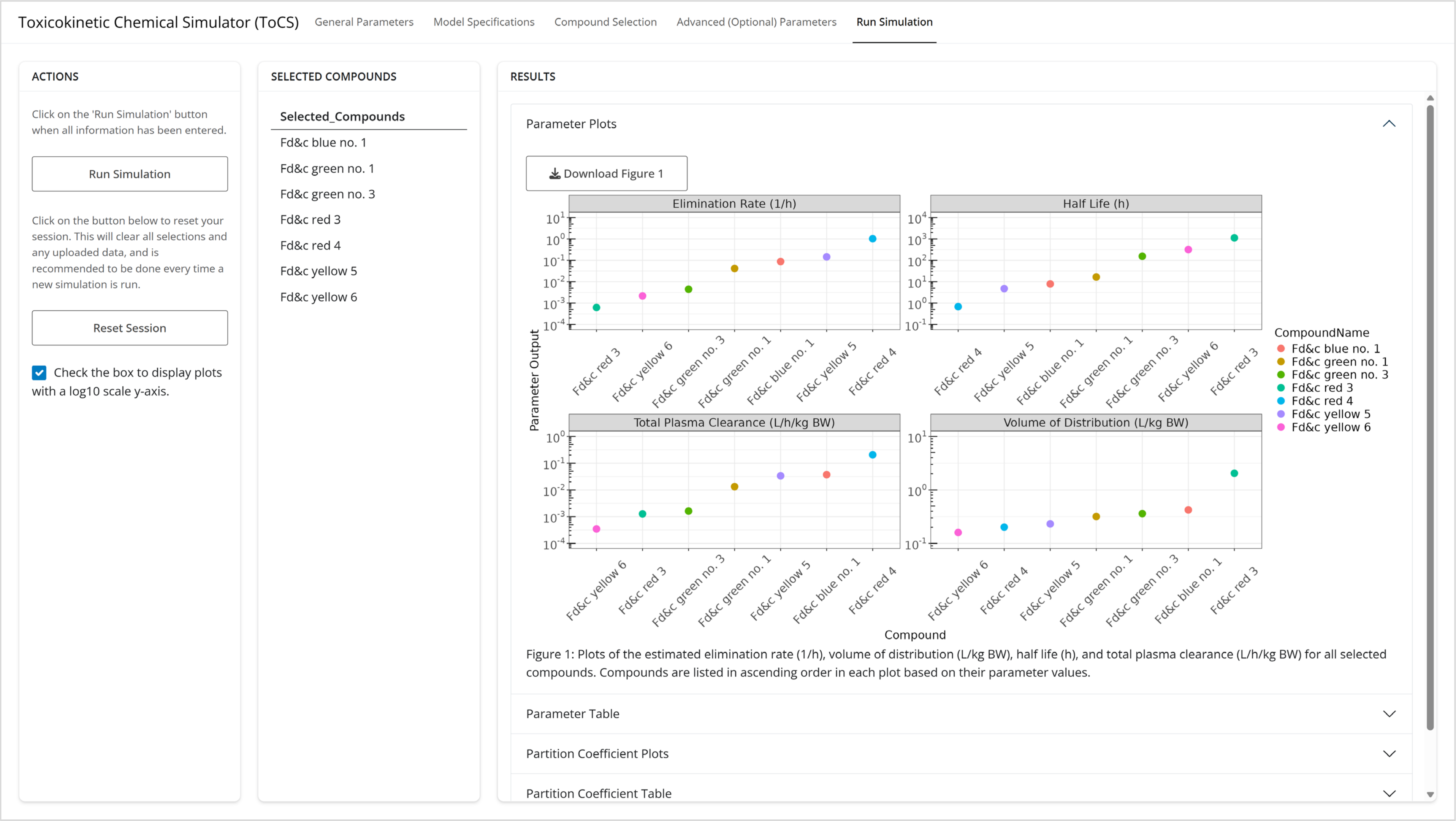


Example 1: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | Run Simulation

ACTIONS

Click on the 'Run Simulation' button when all information has been entered.

Run Simulation

Click on the button below to reset your session. This will clear all selections and any uploaded data, and is recommended to be done every time a new simulation is run.

Reset Session

☑ Check the box to display plots with a log10 scale y-axis.

SELECTED COMPOUNDS

Selected_Compounds
Fd&c blue no. 1
Fd&c green no. 1
Fd&c green no. 3
Fd&c red 3
Fd&c red 4
Fd&c yellow 5
Fd&c yellow 6

RESULTS

Parameter Plots

Parameter Table

Download Table 1

Show 10 entries　　Search:

	CompoundName	EliminationRate	VolumeOfDistribution	HalfLife	TotalClearance
1	Fd&c blue no. 1	0.08791	0.4218	7.885	0.03708
2	Fd&c green no. 1	0.04175	0.3174	16.6	0.01325
3	Fd&c green no. 3	0.004468	0.3606	155.1	0.001611
4	Fd&c red 3	0.0006209	2.047	1116	0.001271
5	Fd&c red 4	1.018	0.2016	0.6809	0.2052
6	Fd&c yellow 5	0.145	0.2311	4.78	0.03352
7	Fd&c yellow 6	0.002178	0.1594	318.2	0.0003471

Showing 1 to 7 of 7 entries　　Previous 1 Next

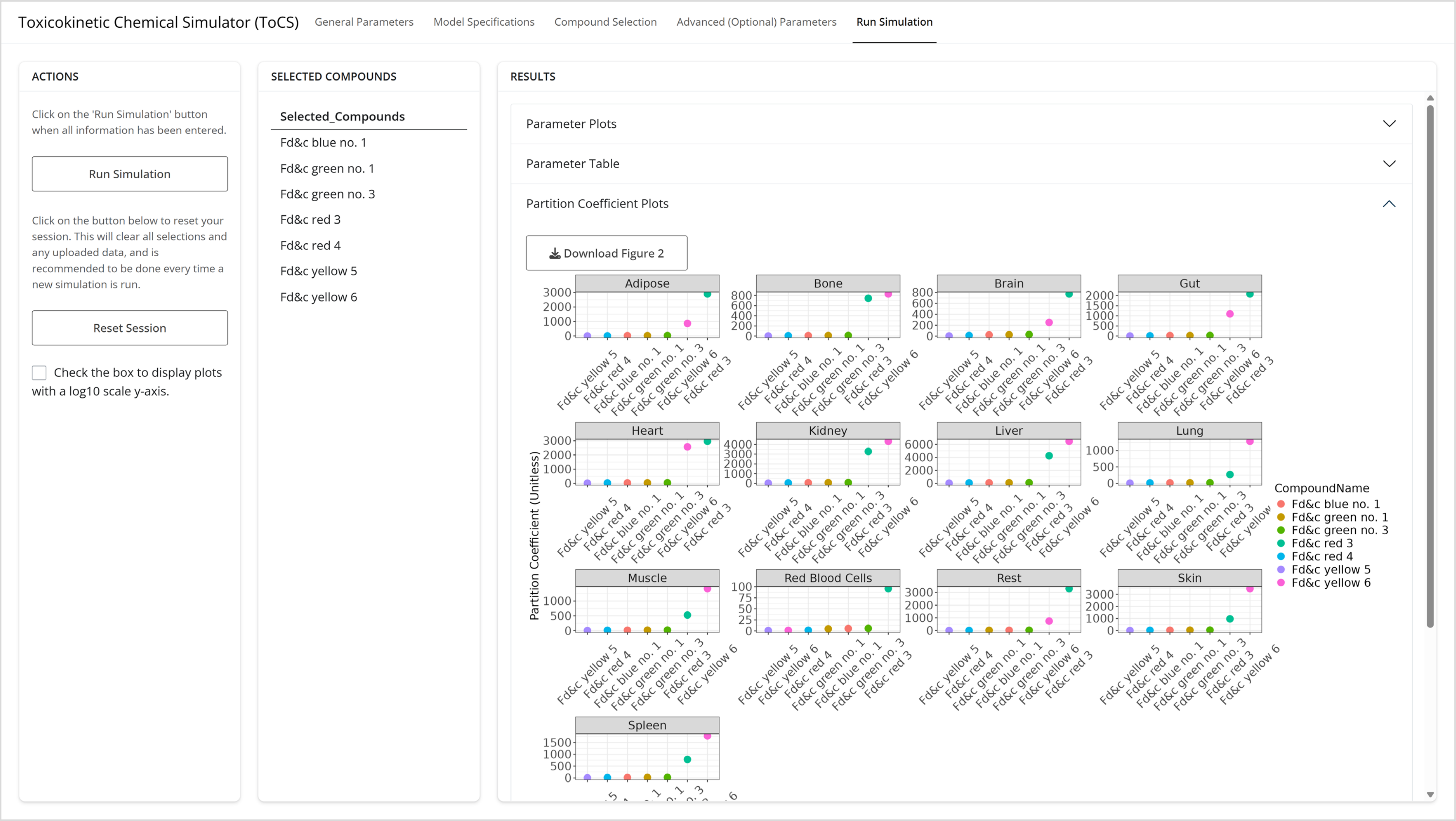
Table 1: Table of estimated elimination rates (1/h), volumes of distribution (L/kg BW), half lifes (h), and total plasma clearances (L/h/kg BW) for all selected compounds.

Partition Coefficient Plots

Partition Coefficient Table

Simulation Parameters

Example 1: Run Simulation Tab



Example 1: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | **Run Simulation**

ACTIONS

Click on the 'Run Simulation' button when all information has been entered.

Run Simulation

Click on the button below to reset your session. This will clear all selections and any uploaded data, and is recommended to be done every time a new simulation is run.

Reset Session

☐ Check the box to display plots with a log10 scale y-axis.

SELECTED COMPOUNDS

Selected_Compounds
Fd&c blue no. 1
Fd&c green no. 1
Fd&c green no. 3
Fd&c red 3
Fd&c red 4
Fd&c yellow 5
Fd&c yellow 6

RESULTS

Partition Coefficient Plots

Download Figure 2

Figure 2: Plots of the estimated partition coefficients (unitless) using Schmitt's method for all tissues available (adipose, bone, brain, gut, heart, kidney,

Example 1: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | **Run Simulation**

ACTIONS

Click on the 'Run Simulation' button when all information has been entered.

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Reset Session

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SELECTED COMPOUNDS

Selected_Compounds
Fd&c blue no. 1
Fd&c green no. 1
Fd&c green no. 3
Fd&c red 3
Fd&c red 4
Fd&c yellow 5
Fd&c yellow 6

RESULTS

Partition Coefficient Plots

Download Figure 2

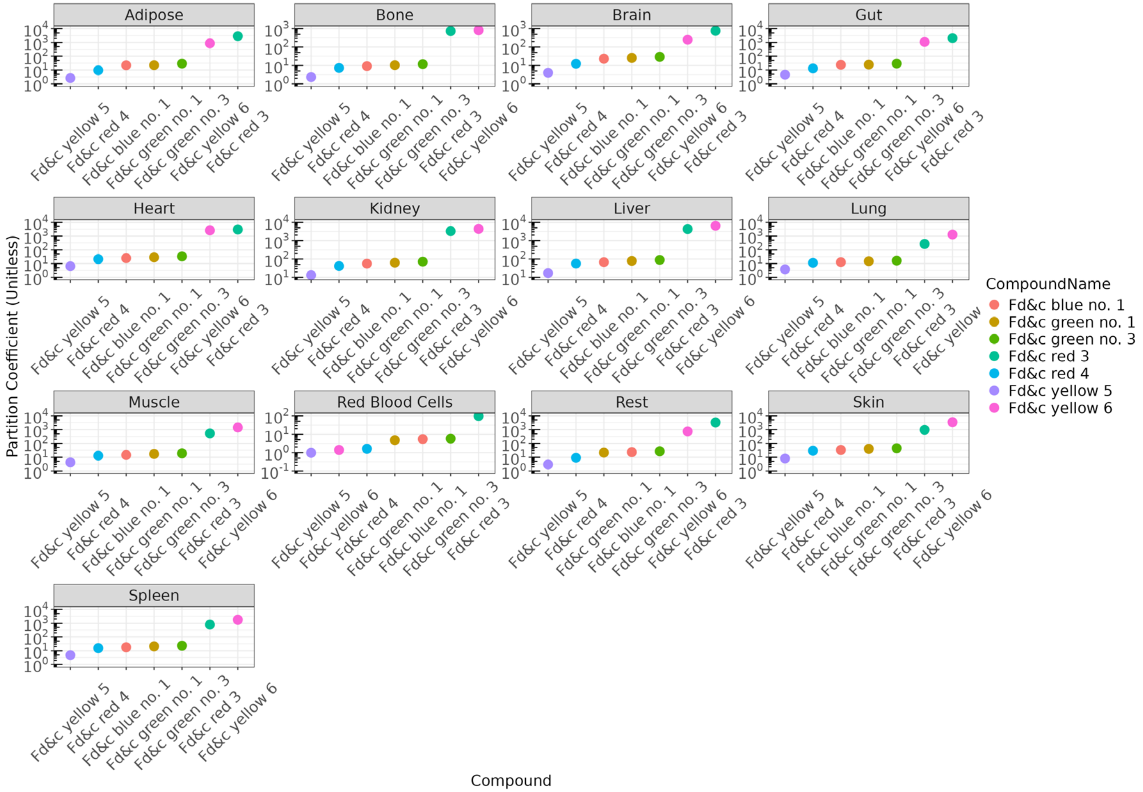

Figure 2: Plots of the estimated partition coefficients (unitless) using Schmitt's method for all tissues available (adipose, bone, brain, gut, heart, kidney,

Example 1: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS) | General Parameters | Model Specifications | Compound Selection | Advanced (Optional) Parameters | **Run Simulation**

ACTIONS

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Click on the button below to reset your session. This will clear all selections and any uploaded data, and is recommended to be done every time a new simulation is run.

Reset Session

☑ Check the box to display plots with a log10 scale y-axis.

SELECTED COMPOUNDS

Selected_Compounds
Fd&c blue no. 1
Fd&c green no. 1
Fd&c green no. 3
Fd&c red 3
Fd&c red 4
Fd&c yellow 5
Fd&c yellow 6

RESULTS

Parameter Plots

Parameter Table

Partition Coefficient Plots

Partition Coefficient Table

Download Table 2

Show 10 entries   Search:

	CompoundName	AdiposePC	BonePC	BrainPC	GutPC	HeartPC	KidneyPC	LiverPC	LungPC	MusclePC	SkinP
1	Fd&c blue no. 1	22.2	9.191	23.11	24.06	24.88	55.94	67.44	12.79	14.41	33
2	Fd&c green no. 1	22.58	10.38	25.63	24.8	28.53	62.97	78.79	14.92	17.06	40
3	Fd&c green no. 3	29.42	11.84	29.37	29.49	33.34	71.85	89.15	16.41	18.85	44
4	Fd&c red 3	2897	744	772.3	2081	2962	3278	4243	269.3	527.3	97
5	Fd&c red 4	9.605	7.327	12.43	13.13	21.22	41.32	56.42	11.41	12.82	29
6	Fd&c yellow 5	2.609	2.316	3.998	4.596	6.445	13.14	16.99	3.831	4.197	8
7	Fd&c yellow 6	868.2	828.8	251.5	1094	2572	4323	6473	1284	1421	34

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Table 2: Table of partition coefficients for all selected compounds in each of the available tissues (adipose, bone, brain, gut, heart, kidney, liver, lung, muscle, skin, spleen, red blood cells (rbc), rest - collective term for remaining tissues).

Simulation Parameters

Download Simulation Parameters